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Multi-Level Security and Cross-Domain Solutions via Homomorphic Policy Evaluation and Zero-Trust Fabric

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Hanzo Team

Hanzo Industries Lux Industries Zoo Labs Foundation
research@hanzo.ai

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Abstract

We present a cross-domain solution (CDS) architecture where data transfer policies are evaluated on encrypted content using fully homomorphic encryption (FHE), ensuring the cross-domain guard never observes plaintext. The architecture integrates four components: (1) a zero-trust overlay network with identity-first micro-segmentation enforcing mandatory access controls at every hop; (2) an FHE policy evaluation engine (ThresholdVM) performing content inspection on ciphertexts using BFV/BGV/CKKS/TFHE schemes with GPU-accelerated NTT operations; (3) a key management system with Shamir-based root key distribution, per-organization encryption isolation, and HIPAA/SOX/SEC compliance controls; and (4) an organization-scoped identity and access management layer deriving security labels from JWT claims and enforcing them at the API gateway. The result is a multi-level security (MLS) architecture where data flows between security domains are cryptographically mediated, no single component has access to both plaintext and policy decisions, and all cross-domain transfers produce immutable audit records anchored to a post-quantum blockchain.

Contents

1 Introduction **4**
1.1 Problem Statement . . . 4
1.2 Our Approach . . . 4

2 Architecture Overview **4**
2.1 Security Invariants . . . 5

3 Zero-Trust Network Fabric **5**
3.1 Identity-First Access Control . . . 5
3.2 Security Label Propagation . . . 5
3.3 Multi-Organization Isolation . . . 5

4 FHE Policy Evaluation **6**
4.1 ThresholdVM: Computation on Encrypted Data . . . 6
4.2 Policy Evaluation Flow . . . 7
4.3 GPU-Accelerated FHE Operations . . . 7
4.4 EVM Precompile Interface . . . 7

5 Key Management for Multi-Level Security **8**
5.1 Hanzo KMS Architecture . . . 8
5.2 Cross-Domain Key Isolation . . . 8
5.3 Compliance Controls . . . 8

6 Audit and Accountability **9**
6.1 Blockchain-Anchored Audit Trail . . . 9
6.2 Attestation Protocol . . . 9

7 Bell-LaPadula Enforcement **9**
7.1 Mandatory Access Control . . . 9
7.2 Implementation . . . 9

8 Comparison with Existing CDS Approaches **10**

9 Performance Considerations **10**
9.1 FHE Overhead . . . 10
9.2 Throughput . . . 10

10 Naval MLS Deployment Scenarios **10**
10.1 Coalition Information Sharing 10
10.2 Analyst Workstation Cross-Domain Access 11
10.3 Shipboard MLS Network 11

11 Conclusion **11**

1 Introduction

Cross-domain solutions enable controlled information sharing between networks operating at different security classifications. Traditional CDS architectures rely on trusted guards that inspect plaintext content against release policies—creating a high-value target where a single compromise exposes all cross-domain traffic.

This paper presents a fundamentally different approach: the guard evaluates policies on *encrypted* content using fully homomorphic encryption. The guard learns only the Boolean policy decision (release/deny), never the content itself. This eliminates the single point of trust while maintaining mandatory policy enforcement.

1.1 Problem Statement

Naval operations require information sharing across multiple security domains:

- **Unclassified** ↔ **Secret**: Intelligence products sanitized for coalition partners.
- **Secret** ↔ **Top Secret/SCI**: Analyst workstations accessing compartmented data.
- **US** ↔ **Coalition**: Multinational operations requiring controlled release.
- **Operational** ↔ **Administrative**: Preventing data spillage between networks.

Current CDS solutions (ISSE Guard, Radiant Mercury, High Speed Guard) process plaintext, requiring the guard itself to be trusted at the highest classification level. A compromised guard exposes all cross-domain traffic.

1.2 Our Approach

1. Data encrypted at source under FHE public key.
2. Guard evaluates content policy homomorphically on ciphertext.
3. Encrypted Boolean result threshold-decrypted by MPC committee.
4. If approved, data re-encrypted for destination domain.
5. Guard never sees plaintext; committee never sees policy result.

2 Architecture Overview

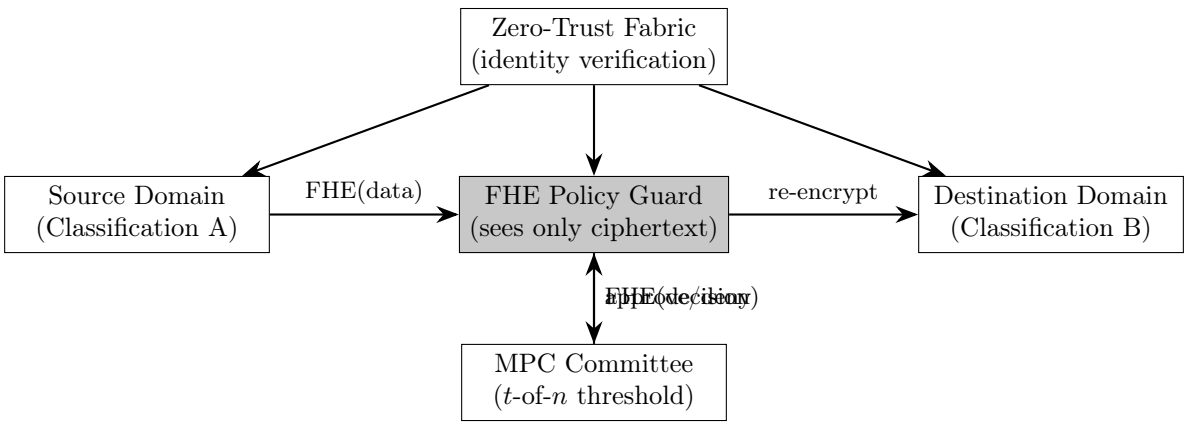

Figure 1: FHE-based cross-domain solution architecture.

2.1 Security Invariants

1. **Guard blindness**: The policy guard processes only ciphertexts. It cannot distinguish encrypted classified data from encrypted noise.
2. **Committee separation**: MPC committee members see decryption shares, never the full policy result or the data. Threshold (t-of-n) prevents minority collusion.
3. **Mandatory enforcement**: Zero-trust fabric verifies cryptographic identity at every hop. Security labels derived from JWT claims cannot be forged.
4. **Audit immutability**: Every cross-domain transfer produces a blockchain-anchored receipt with PQ signatures.

3 Zero-Trust Network Fabric

3.1 Identity-First Access Control

Hanzo ZT provides the network underlay with mandatory access controls:

- **Cryptographic identity**: Every endpoint possesses a unique X.509 certificate with HSM-backed private key (PKCS#11, YubiHSM).
- **No implicit trust**: Network position (IP address, VLAN, subnet) confers zero access. Every connection requires identity verification.
- **Micro-segmentation**: Policies define which identities can communicate. Cross-domain traffic flows only through designated guard endpoints.
- **Policy enforcement**: Applied at every hop in the overlay mesh, not just at perimeter firewalls.

3.2 Security Label Propagation

Security labels are derived from the IAM identity system:

Listing 1: Security label derivation from JWT claims.

```
// Gateway extracts identity from JWT
X-Org-Id:    "navy-secret"     // Organization = domain
X-User-Id:   "analyst-0042"    // Subject identity
X-User-Email: "j.doe@navy.mil" // Attribution

// Gateway enforces: only "navy-secret" org members
// can reach services in "secret" namespace
// Cross-domain requires FHE guard intermediary
```

The API gateway strips all client-supplied identity headers and derives them exclusively from cryptographically verified JWT claims. This prevents label forgery.

3.3 Multi-Organization Isolation

The platform supports 4+ organizations with complete cryptographic isolation:

Organization	Root Key	Isolation Mechanism
hanzo	Separate AES-256	KMS per-org encryption
lux	Separate AES-256	Separate key hierarchy
zoo	Separate AES-256	Org-scoped IAM policies
pars	Separate AES-256	Independent audit logs

Table 1: Per-organization cryptographic isolation.

Each organization has its own root encryption key, IAM policies, KMS secrets, and audit trail. Cross-organization data access requires explicit policy and traverses the FHE guard.

4 FHE Policy Evaluation

4.1 ThresholdVM: Computation on Encrypted Data

ThresholdVM operates on the Lux T-Chain, providing FHE as a native virtual machine:

Scheme	Domain	CDS Application
BFV	Integer	Keyword matching on encrypted text
BGV	Modular	Classification marker detection
CKKS	Approx. real	ML-based content classification
TFHE	Boolean	Arbitrary policy Boolean circuits

Table 2: FHE schemes mapped to CDS operations.

4.2 Policy Evaluation Flow

Algorithm 1 FHE Cross-Domain Policy Evaluation

Source encrypts data: $c \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{data})$
Source encrypts metadata: $m \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{labels})$
Source sends $(c, m, \text{dest_domain})$ to Guard

Guard evaluates policy homomorphically:
  $r \leftarrow \text{FHE.Eval}(\text{policy_circuit}, c, m, \text{dest_domain})$
  (Guard sees only ciphertexts; r is an encrypted Boolean)

Guard sends r to MPC committee for threshold decryption
for each committee member $i = 1, \ldots, n$ **do**
  $d_i \leftarrow \text{ThresholdDecrypt}(\text{sk}_i, r)$
end for
decision $\leftarrow \text{Reconstruct}(d_1, \ldots, d_t)$ (t shares suffice)

if decision = APPROVE **then**
  Re-encrypt data for destination domain key
  Forward to destination via zero-trust fabric
  Anchor audit receipt to blockchain (ML-DSA-65 signed)
else
  Log denial (encrypted), destroy ciphertext
end if

4.3 GPU-Accelerated FHE Operations

LuxCPP provides Metal and CUDA kernels for FHE primitives:

- **NTT/iNTT**: Number Theoretic Transform for polynomial multiplication in $\mathbb{Z}_q[X]/(X^n + 1)$.
- **TFHE bootstrap**: Programmable bootstrap for refreshing ciphertext noise.
- **Blind rotate**: Key operation in TFHE gate evaluation.
- **Batch operations**: Parallel ciphertext processing across GPU threads.

GPU acceleration reduces FHE policy evaluation from seconds to milliseconds, making real-time cross-domain filtering practical.

4.4 EVM Precompile Interface

Smart contracts can invoke FHE operations via precompiles:

Address	Function	Gas
`0x0700`	FHE encrypt/decrypt/eval	500,000
`0x0701`	Ciphertext ACL management	25,000
`0x0702`	Input proof verification	50,000
`0x0703`	Cross-chain gateway	100,000

Table 3: FHE EVM precompiles for on-chain policy enforcement.

5 Key Management for Multi-Level Security

5.1 Hanzo KMS Architecture

The KMS provides 8 subsystems supporting MLS key lifecycle:

1. **Shamir Secret Sharing**: Root key split into n shares over $GF(2^8)$; t required for reconstruction. Custodians geographically distributed.
2. **Transit Engine**: Encryption-as-a-service with key versioning. Each security domain has independent transit keys with separate rotation schedules.
3. **HPKE Key Wrapping**: Content encryption keys (CEKs) wrapped using hybrid HPKE (X25519 + ML-KEM-768) per RFC 9180.
4. **Seal/Unseal Barrier**: AES-256-GCM barrier protecting all stored secrets. Auto-unseal via AWS/GCP KMS for operational environments.
5. **ACL Policy Engine**: Path-based capabilities with template variables (`{{identity.org_id}}`) enabling per-domain access rules.
6. **Token Hierarchy**: Service, batch, and recovery tokens with parent-child revocation cascade. Compromised token revocation propagates instantly.
7. **Dynamic Secrets**: Temporary database credentials (PostgreSQL, MySQL, Redis, MongoDB) scoped to security domain and auto-expired.
8. **TFHE-KMS Bridge**: FHE key management integrated with policy evaluation pipeline.

5.2 Cross-Domain Key Isolation

- Each security domain has an independent root key, never shared.
- Cross-domain data transfer requires re-encryption under destination domain key.
- Re-encryption occurs only after FHE policy approval and threshold decryption.
- No single KMS instance holds keys for multiple domains.

5.3 Compliance Controls

Standard	Implementation
HIPAA	Break-glass tokens (time-limited), WORM audit logs
SEC 17a-4	6–7 year retention, immutable SHA-256 chain
SOX	Regulator escrow shards, cooperative reconstruction
GDPR	Per-org keys, graceful revocation
CNSA 2.0	PQ algorithms for all key operations

Table 4: Regulatory compliance controls in KMS.

6 Audit and Accountability

6.1 Blockchain-Anchored Audit Trail

Every cross-domain transfer produces an immutable audit record:

- **Record contents**: Source domain, destination domain, policy ID, decision (approve/-deny), timestamp, data hash (not data), requestor DID.
- **Signature**: ML-DSA-65 (post-quantum) ensuring long-term non-repudiation.
- **Anchoring**: Record hash committed to Lux blockchain via Quasar consensus.
- **WORM guarantee**: Write-once-read-many with SHA-256 hash chain prevents retroactive modification.

6.2 Attestation Protocol

Domain-typed attestations with equivocation detection:

- **CrossDomainTransfer**: Records each approved transfer with full provenance.
- **PolicyUpdate**: Records changes to cross-domain policies.
- **KeyRotation**: Records domain key rotation events.
- Signing contradictory attestations (equivocation) is automatically detected and flagged.

7 Bell-LaPadula Enforcement

7.1 Mandatory Access Control

The architecture enforces Bell-LaPadula properties:

Definition 7.1 (Simple Security Property (No Read Up))**.** A subject at security level L_s cannot read an object at security level L_o where $L_o > L_s$. Enforced by the zero-trust fabric: identity JWT contains org-scoped level; gateway rejects requests to higher-level services.

Definition 7.2 (*-Property (No Write Down))**.** A subject at security level L_s cannot write to an object at security level L_o where $L_o < L_s$. Enforced by the FHE guard: data flowing from high to low domain must pass policy evaluation.

7.2 Implementation

1. **Read control**: Gateway checks `X-Org-Id` (derived from JWT) against service security label. Requests to higher domains rejected before reaching backend.
2. **Write control**: All data egress from a domain passes through the FHE guard. Downward flows require policy approval; upward flows are unrestricted.
3. **Label integrity**: Security labels are gateway-derived from cryptographic JWT claims. Client-supplied labels are stripped. Labels cannot be forged without compromising the IAM signing key.

8 Comparison with Existing CDS Approaches

Feature	**Traditional Guard**	**Data Diode**	**FHE Guard (Ours)**
Plaintext exposure	Guard sees all	One-way only	Never
Bidirectional	Yes	No	Yes
Content inspection	Full	None	On ciphertext
Single point of trust	Guard	No	No (threshold)
PQ-safe	Depends	N/A	Yes (FIPS)
Audit	Guard logs	Physical	Blockchain
ML classification	On plaintext	N/A	On ciphertext

Table 5: CDS approach comparison.

The FHE-based approach uniquely provides bidirectional content-inspected cross-domain transfer with zero plaintext exposure. No commercially available CDS offers this capability.

9 Performance Considerations

9.1 FHE Overhead

Operation	**CPU**	**GPU (Metal)**
TFHE gate evaluation	13 ms	0.8 ms
BFV keyword match	45 ms	3.2 ms
CKKS ML inference	200 ms	15 ms
Threshold decrypt (t=3)	50 ms	12 ms
Total policy eval	**∼300 ms**	**∼30 ms**

Table 6: FHE policy evaluation latency.

GPU acceleration brings FHE policy evaluation to 30 ms—practical for real-time cross-domain filtering of message traffic, though not suitable for bulk file transfer without parallelization.

9.2 Throughput

With GPU acceleration and pipelined evaluation:

- **Messages**: ∼33 messages/second per GPU (real-time chat feasible).
- **Bulk data**: Parallelize across multiple GPUs; 8-GPU node achieves ∼264 evaluations/second.
- **Ciphertext expansion**: 100–1000× plaintext size (FHE-inherent). ZAP zero-copy design mitigates the internal memory cost.

10 Naval MLS Deployment Scenarios

10.1 Coalition Information Sharing

- US Secret data evaluated against release policy for Five Eyes partners.

- FHE guard checks classification markers, source restrictions, and content sensitivity—all on ciphertext.
- Approved data re-encrypted under coalition domain key and forwarded.
- Full audit trail anchored to blockchain for post-operation review.

10.2 Analyst Workstation Cross-Domain Access

- Analyst at Secret level queries Top Secret/SCI database.
- Query encrypted under FHE key; database returns encrypted results.
- FHE guard evaluates sanitization policy on encrypted response.
- Approved (sanitized) response decrypted for analyst's domain.

10.3 Shipboard MLS Network

- Multiple security domains on single shipboard network (physically separated VLANs, logically separated via zero-trust overlay).
- FHE guards at domain boundaries, GPU-accelerated for real-time messaging.
- KMS with separate root keys per domain, HSM-protected.
- All cross-domain traffic audited with PQ-signed blockchain receipts.

11 Conclusion

We have presented a cross-domain solution architecture where the guard evaluates policies on encrypted content, eliminating the traditional single point of trust. The combination of FHE policy evaluation, zero-trust network fabric, threshold MPC decryption, and blockchain-anchored audit provides a multi-level security architecture with no plaintext exposure at the guard, no single point of key compromise, and immutable accountability.

This approach represents a paradigm shift in cross-domain security: instead of trusting the guard to handle plaintext responsibly, we make it cryptographically impossible for the guard to access plaintext at all.

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